Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

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Supplementary Information

Table S1: Details of stages of Heating using NVT Ensemble

Stage	**Temperature range (K)**	**Time (ns)**
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
Stage 4	300 to 300	7.5 to 10

Table S2: Parameters used for pocket identification in POVME2

Parameter	**Value**
pocket detection resolution	4.0 Å
pocket measuring resolution	1.0 Å
clashing cutoff	3.5 Å
number of neighbors	4
number of spheres	5
sphere padding	5.0 Å

Table S3: Results of Pocket Detection using Fpocket

Pocket	**Fpocket Score**	**Druggability Score**	**Hydrophobicity Score**
Site P1	0.13	0.001	46.38
Site P2	0.051	0.002	10.89
Site P3	0.076	0.02	27.5
Site P4	0.11	0.005	37.33

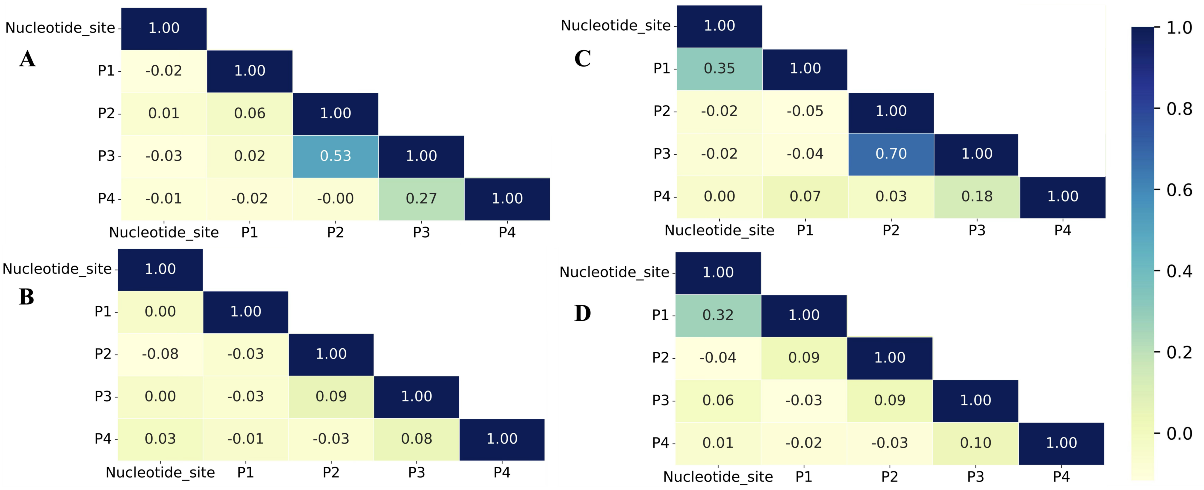

Figure S1: Pearson's Correlation for the volume fluctuations among different pockets and the effect of nucleotide binding. **(A)** GDP-bound Rac1; **(B)** GTP-bound Rac1; **(C)** Nucleotide-free state created by deleting GDP from Rac1; **(D)** Nucleotide-free state created by deleting GTP from Rac1.

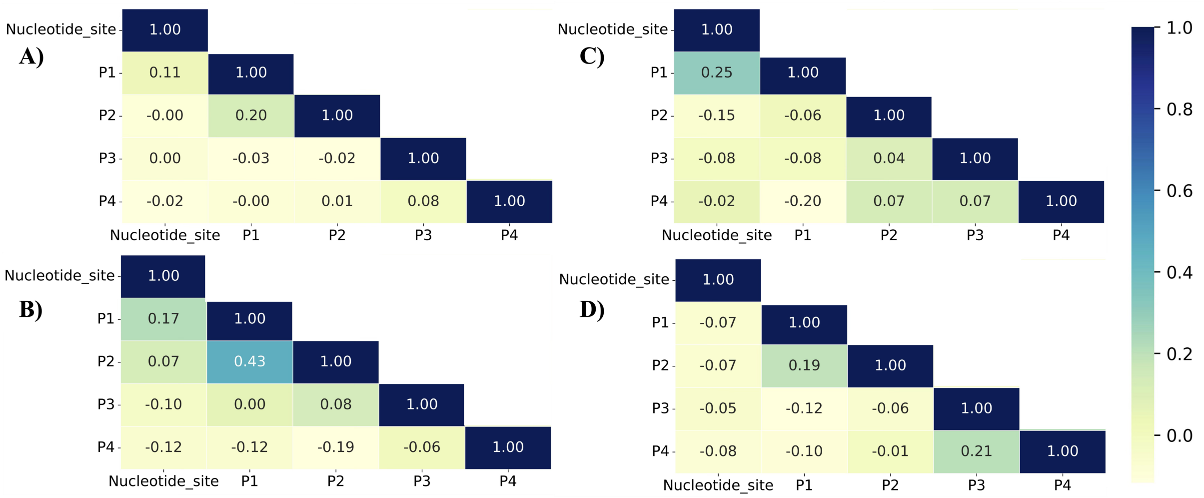

Figure S2: Pearson's Correlation for the volume fluctuations among different pockets and the effect of GEF binding. **(A)** PREX1-bound Rac1; **(B)** TIAM1-bound Rac1; **(C)** VAV1-bound Rac1; **(D)** DOCK2-bound Rac1. Rac1 in all GEF-bound systems is in the nucleotide-free state.

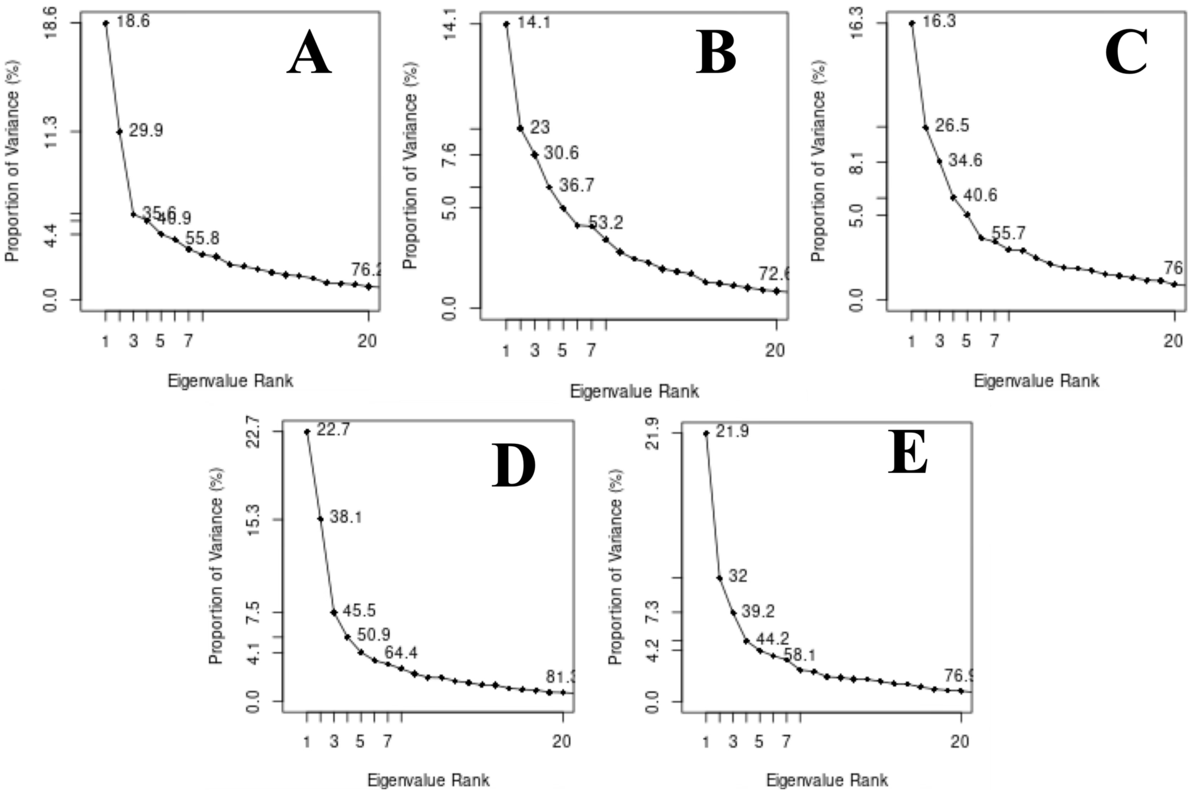

Figure S3: Scree plot to compute the cumulative variance explained by principal components. **(A)** *apo*-Rac1 in TIP3P water box; **(B)** GTP-bound Rac1 in TIP3P water box; **(C)** *apo*-Rac1 in TIP3P water+isopropanol box; **(D)** *apo*-Rac1 in TIP3P water+phenol box; **(E)** *apo*-Rac1 in TIP3P water+toluene box.

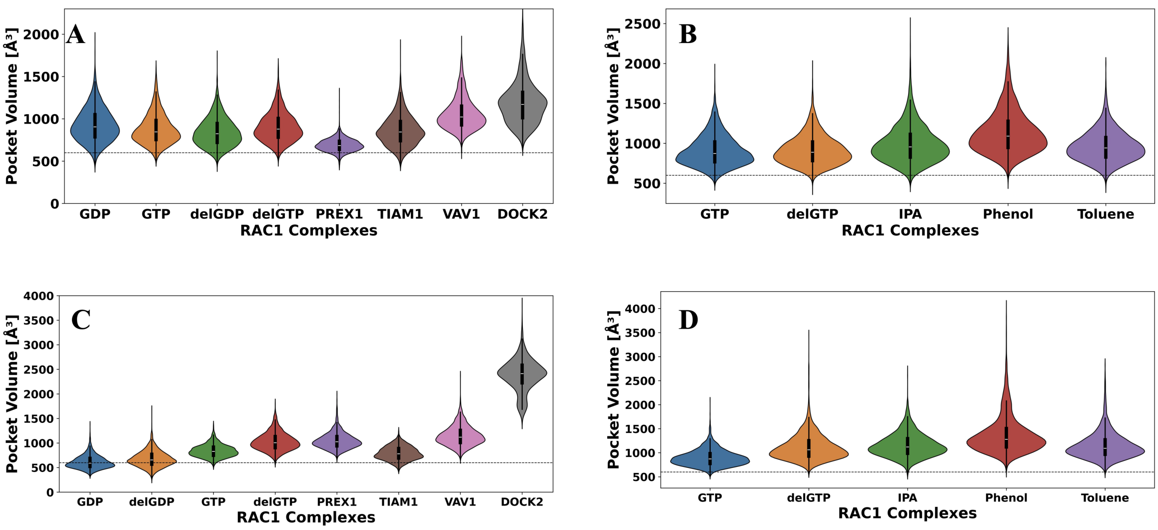

Figure S4: Analysis of pocket volume changes by combining pockets P1 with P2, and P3 with P4. **(A)** Combined pocket volume analysis for site P1+P2 ; **(B)** Combined pocket volume analysis for site P1+P2 in co-solvent; **(C)** Combined pocket volume analysis for site P3+P4; **(D)** Combined pocket volume analysis for site P3+P4 in co-solvent. The threshold value, beyond which a pocket may be considered amenable for small molecule binding, was set according to the following main text references[36–38] and indicated by the dashed line.

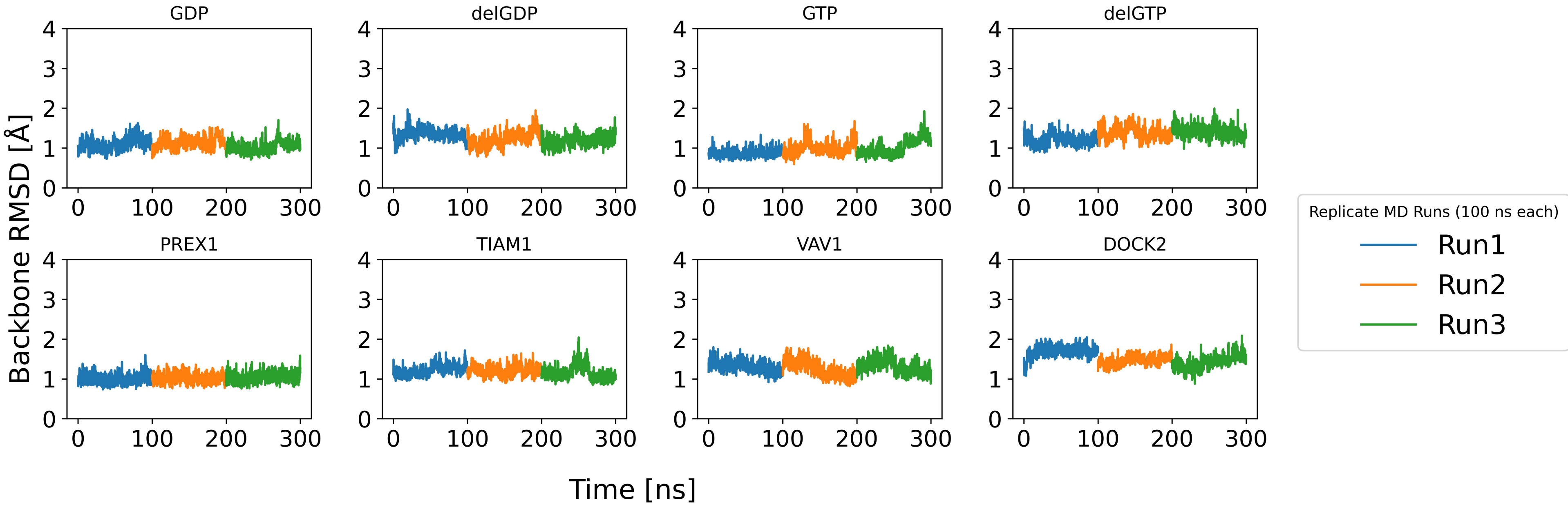

Figure S5: Time-evolution of RMSD of backbone atoms from Full Rac1 protein.

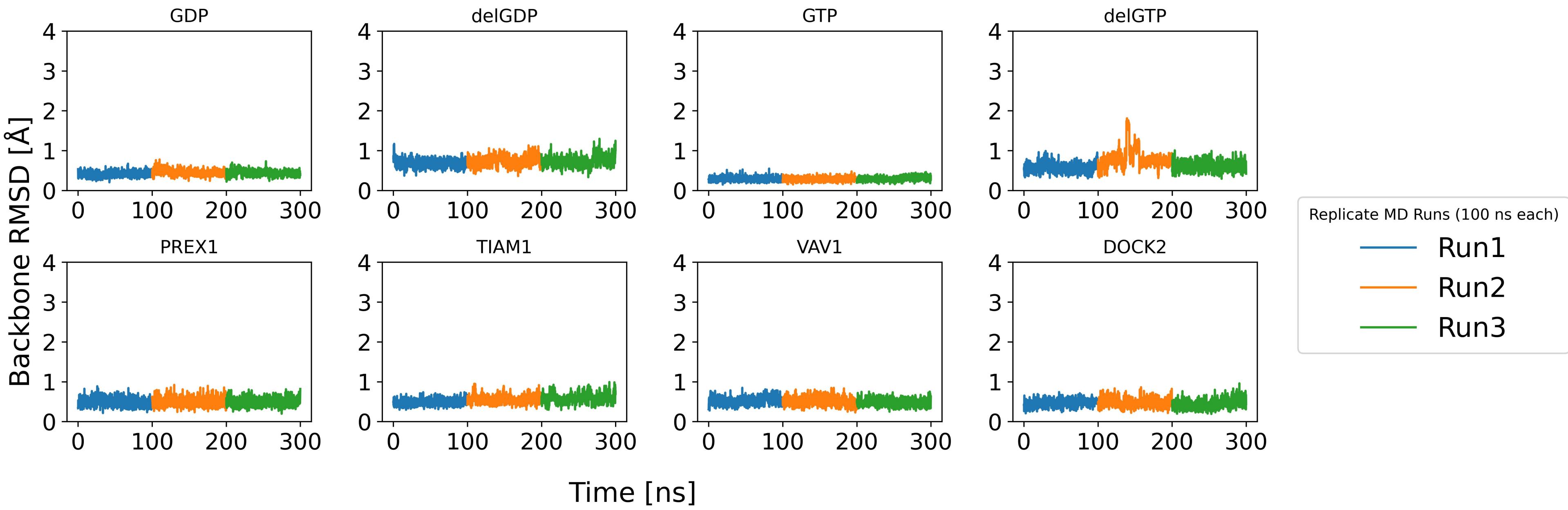

Figure S6: Time-evolution of RMSD of backbone atoms from the P-loop region.

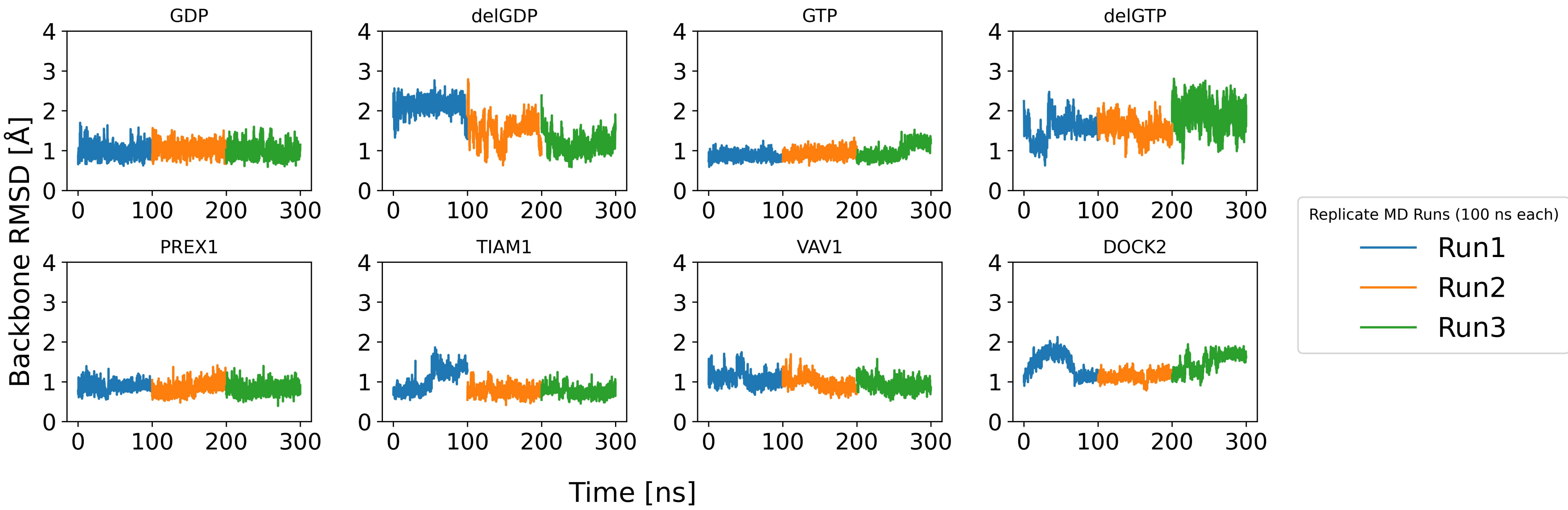

Figure S7: Time-evolution of RMSD of backbone atoms from the switch I region.

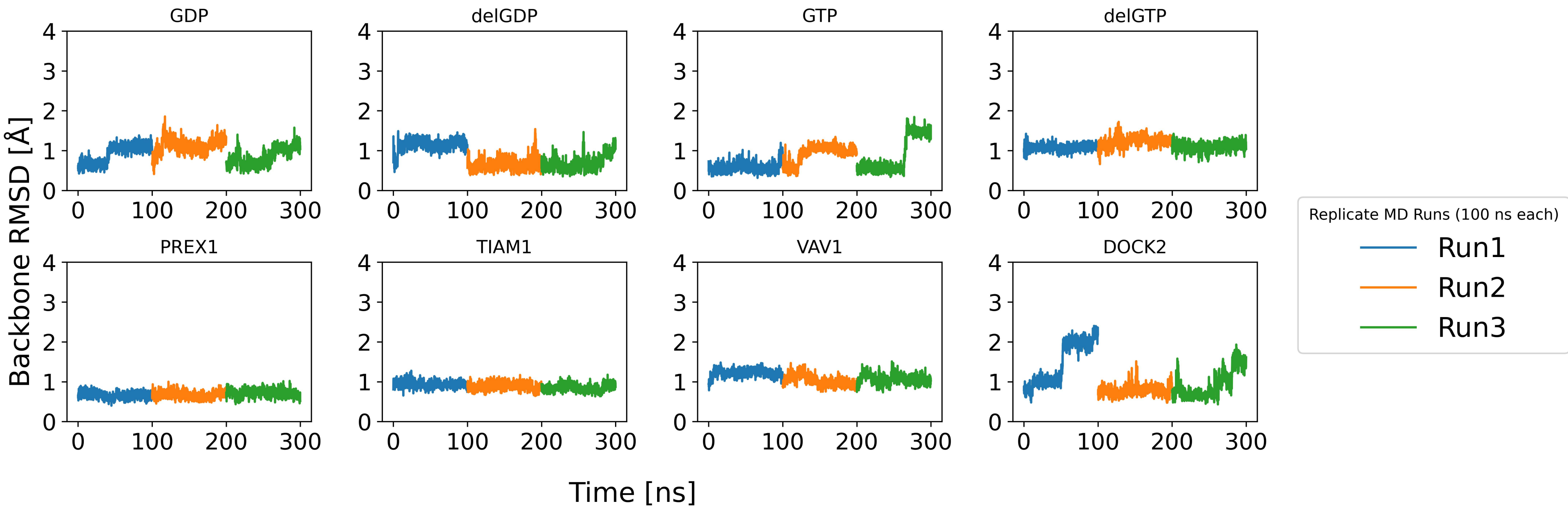

Figure S8: Time-evolution of RMSD of backbone atoms from the switch II region.

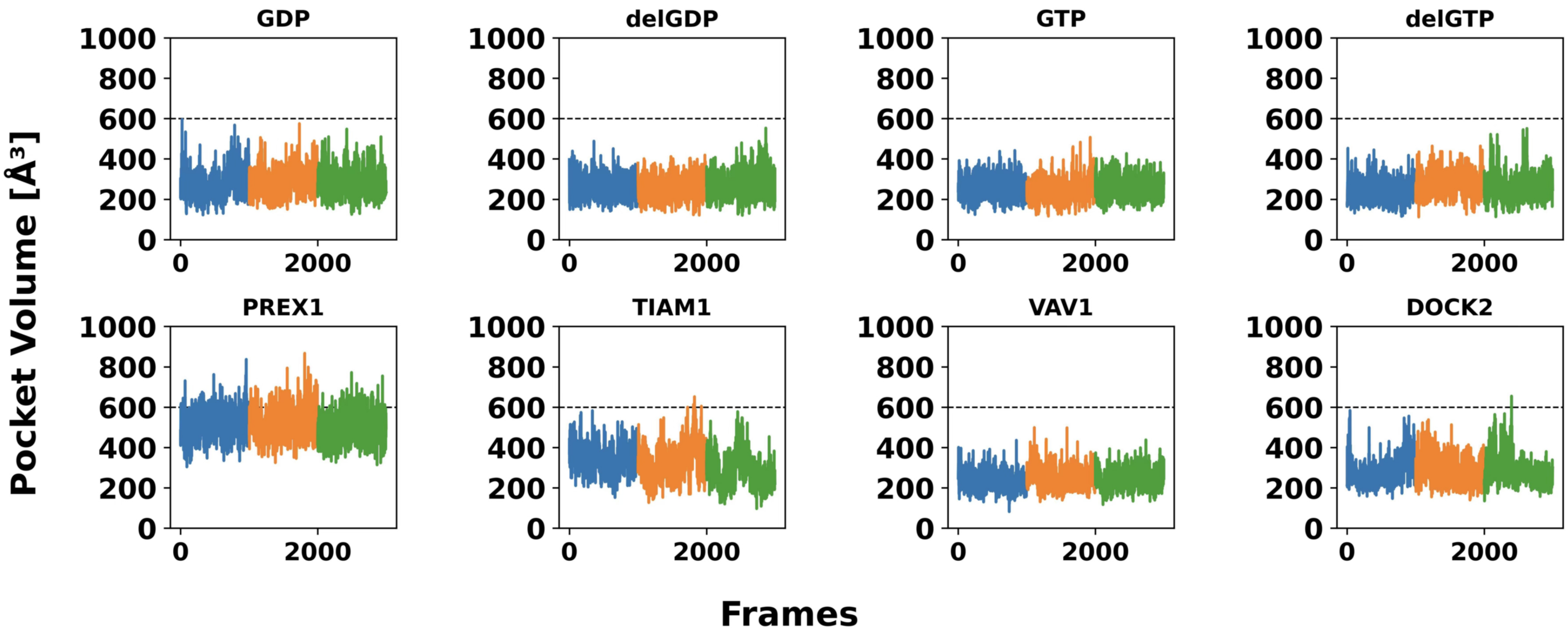

Figure S9: Time-evolution of pocket volume for Site P1.

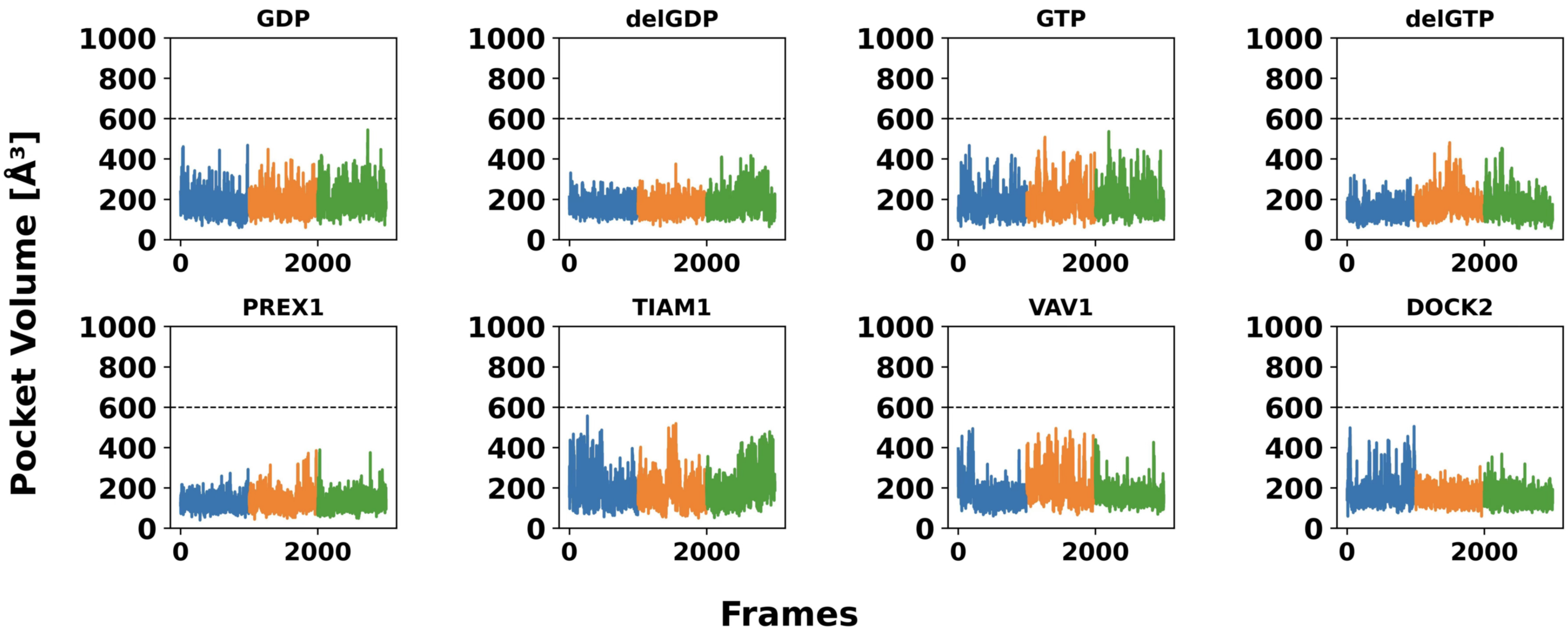

Figure S10: Time-evolution of pocket volume for Site P2.

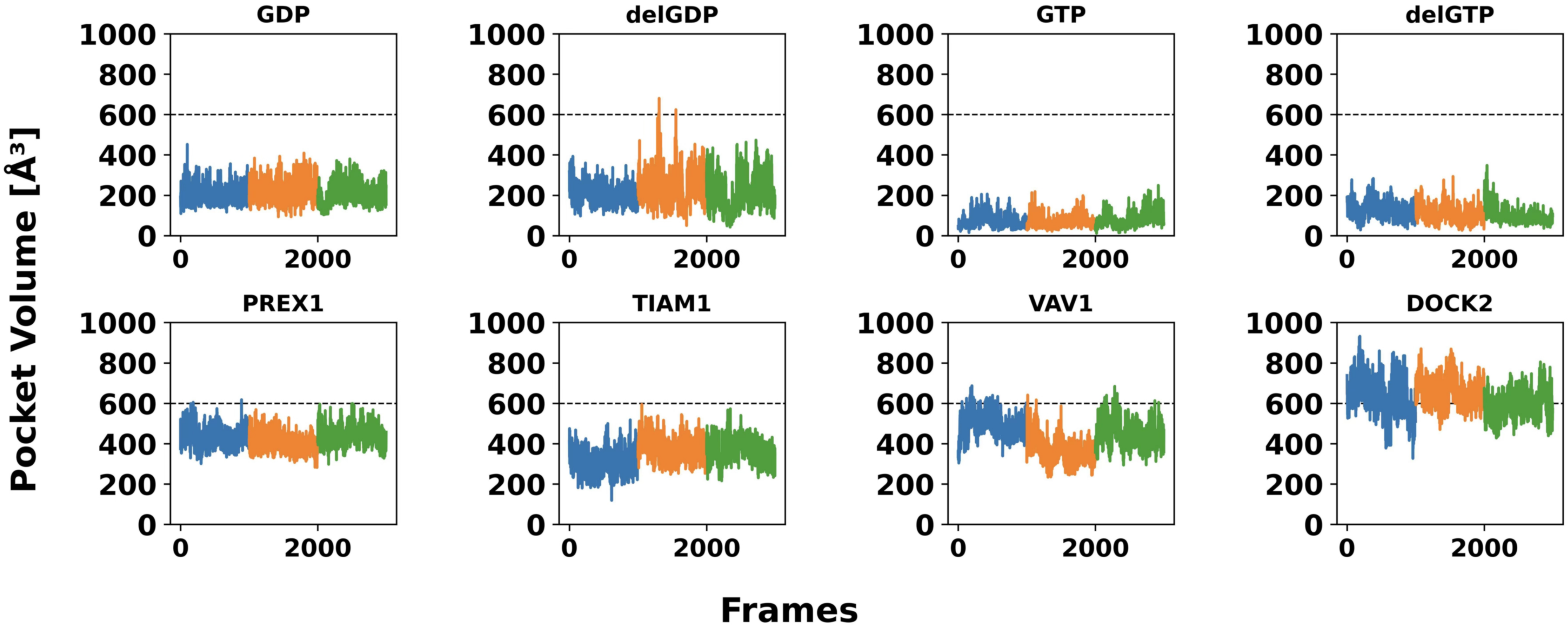

Figure S11: Time-evolution of pocket volume for Site P3.

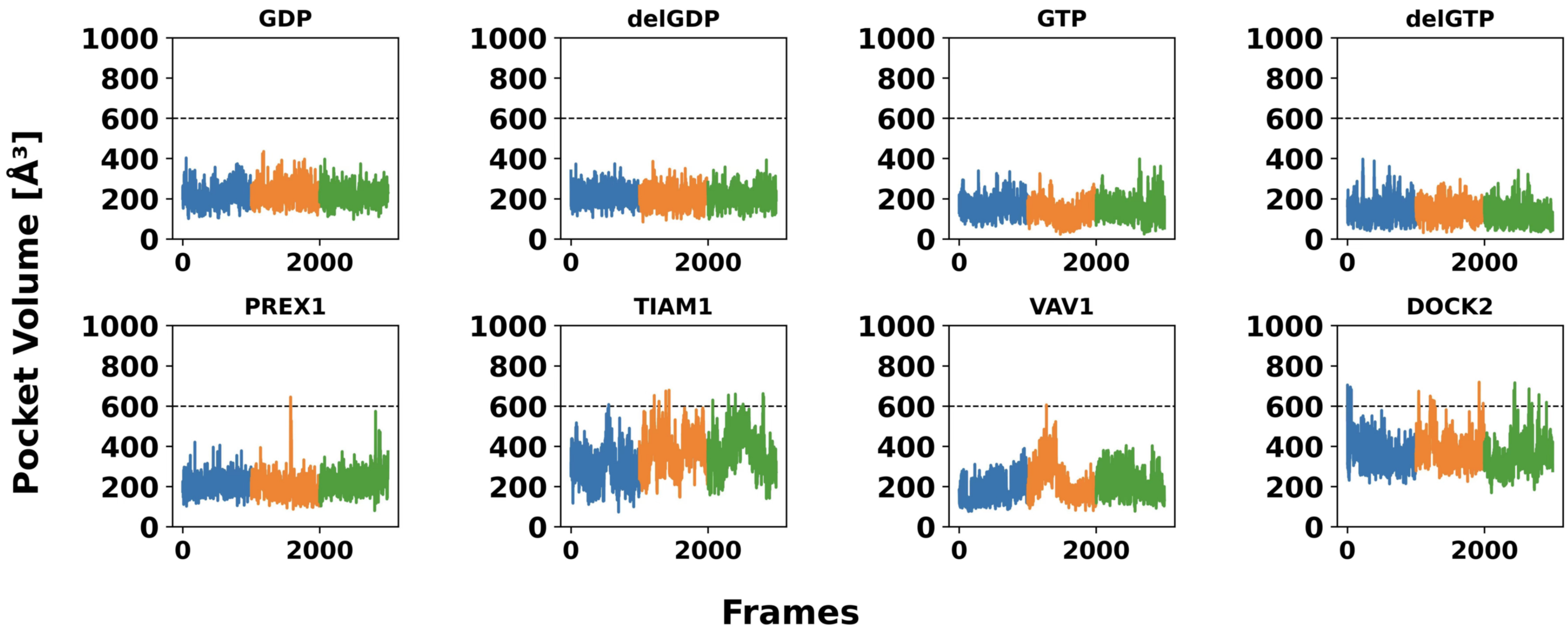

Figure S12: Time-evolution of pocket volume for Site P4.

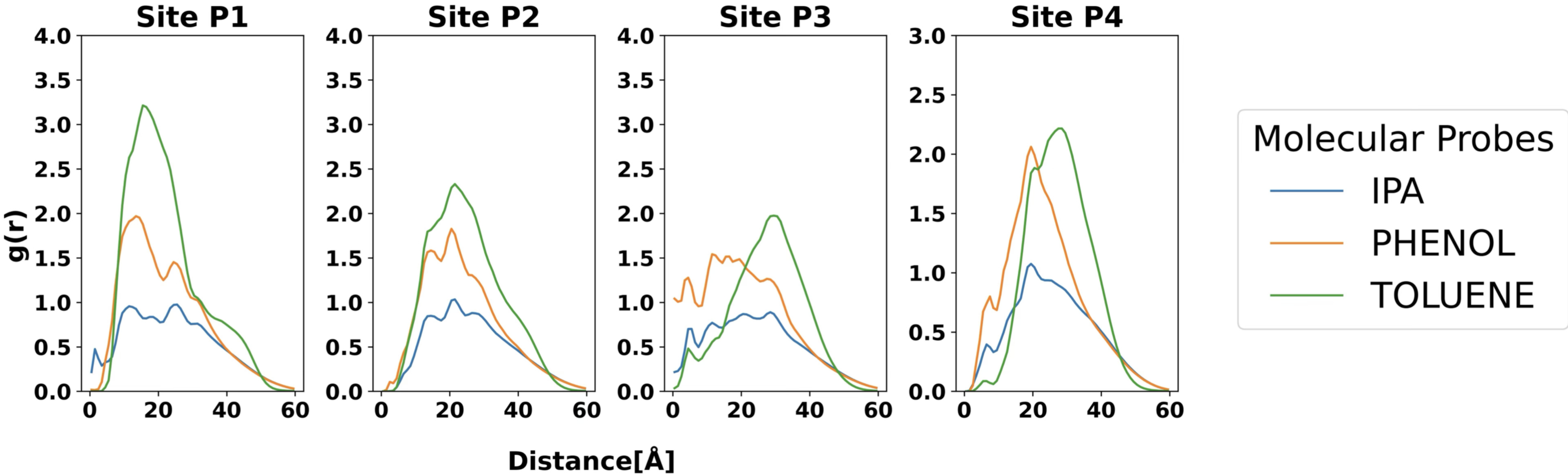

Figure S13: Radial Distribution Function of the Molecular Probes computed for each of the new pockets identifoed in this work.

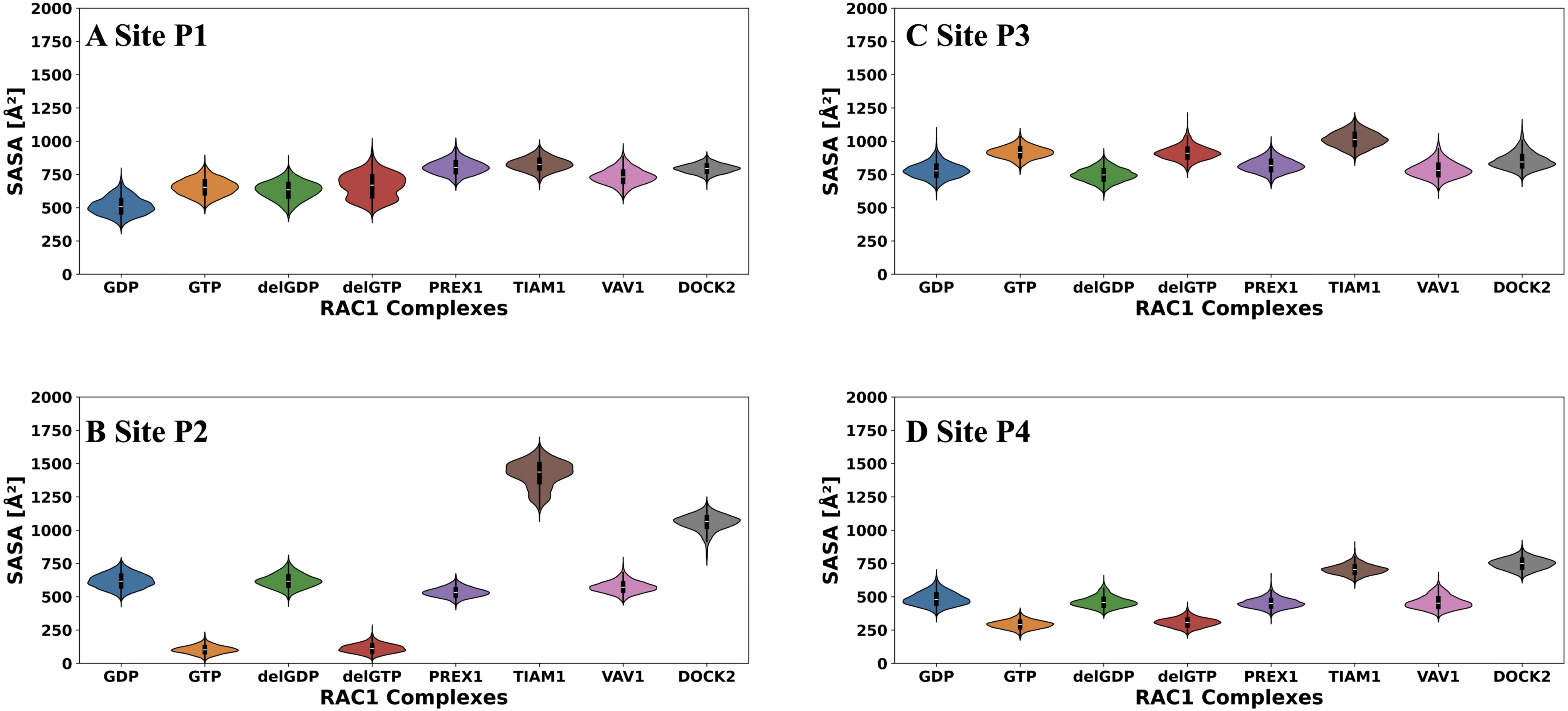

Figure S14: Analysis of the Solvent-Accessible Surface Area using a 1.4 Å probe accross all the newly identified site to understand the effect of nucleotide and GEF binding

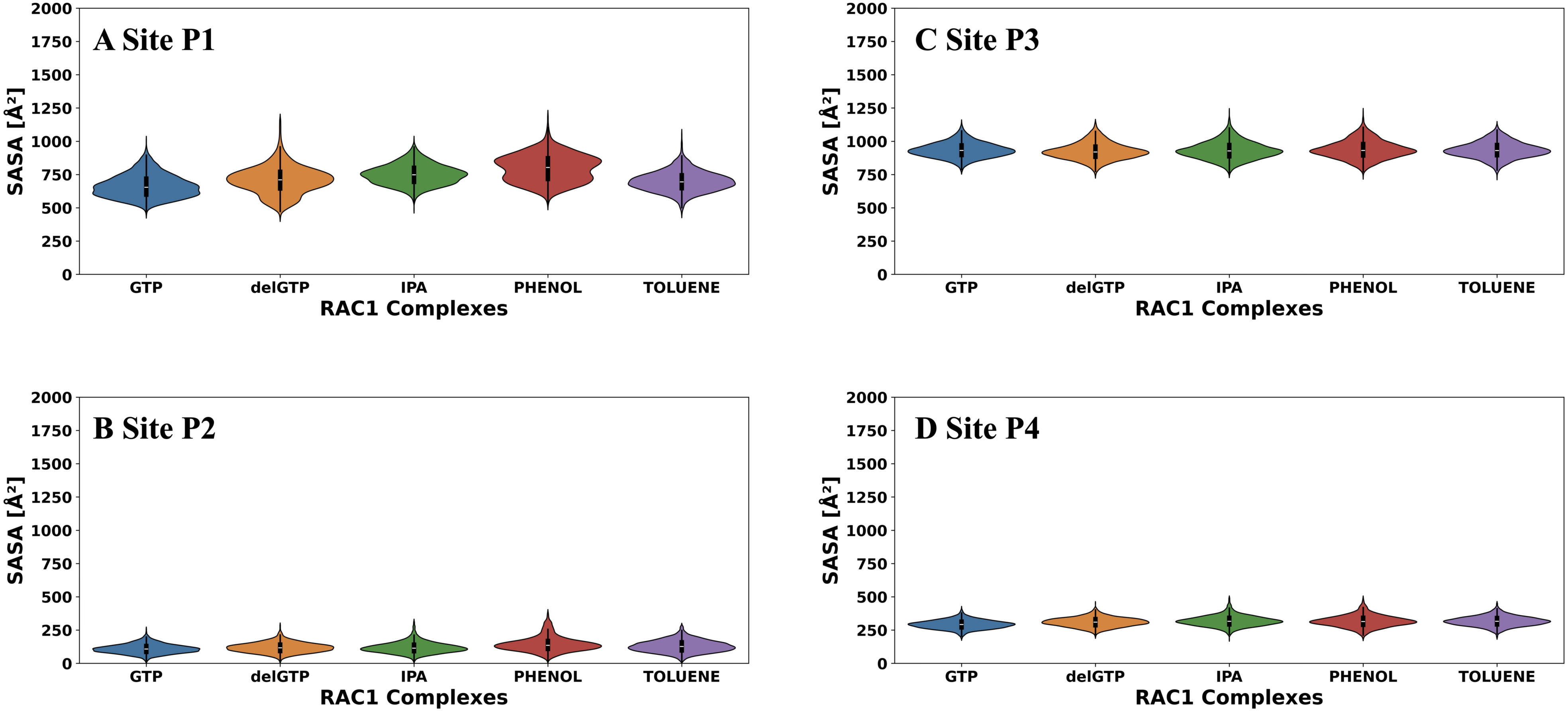

Figure S15: Analysis of the Solvent-Accessible Surface Area using a 1.4 Å probe accross all the newly identified site to understand the effect of Molecular probes